

6.1 Uniform distribution.

A probability distribution is called uniform if the distribution density function remains constant.

Let us find the uniform distribution density function $f(x)$ on the segment $[a,b]$

$f(x)$ remains constant, i.e. $f(x)=C$.

Let's find the constant C from the properties of the distribution function

$$\int_a^b f(x)dx = 1.$$

Or

$$\int_a^b Cdx = 1,$$

$$Cx \Big|_a^b = 1, \quad C(b - a) = 1.$$

From here

$$C = \frac{1}{b - a}.$$

Thus, a continuous random variable X is called uniformly distributed over the interval $[a, b]$

if its probability density has the form:

$$f(x) = \begin{cases} 0, & \text{at } x < a; \\ \frac{1}{b-a}, & \text{at } a \leq x \leq b; \\ 0, & \text{at } x > b. \end{cases} \quad (6.1)$$

The probability distribution function of a uniformly distributed random variable X has the form

$$F(x) = \begin{cases} 0, & \text{at } x < 0; \\ \frac{x-a}{b-a}, & \text{at } a \leq x \leq b; \\ 0, & \text{at } x > b. \end{cases} \quad (6.2)$$

Numerical characteristics for uniform distribution:

a) Expected value

$$M(X) = \int_a^b xf(x)dx,$$

$$M(X) = \int_a^b x \frac{1}{b-a} dx = \frac{1}{b-a} \frac{x^2}{2} \Big|_a^b =$$

$$= \frac{b^2 - a^2}{2(b-a)} = \frac{(b-a)(b+a)}{2(b-a)} = \frac{b+a}{2}.$$

b) Dispersion

$$D(X) = \int_a^b x^2 f(x) dx - [M(X)]^2,$$

we get

$$D(X) = \int_a^b x^2 \frac{1}{b-a} dx - \left[\frac{b+a}{2} \right]^2 = \frac{1}{b-a} \frac{x^3}{3} \Big|_a^b - \frac{(a+b)^2}{4} =$$

$$= \frac{b^3 - a^3}{3(b-a)} - \frac{(a+b)^2}{4} = \frac{(b-a)(b^2 + ab + a^2)}{3(b-a)} - \frac{(a+b)^2}{4} =$$

$$= \frac{(b^2 + ab + a^2)}{3} - \frac{(a+b)^2}{4} =$$

$$= \frac{4b^2 + 4ab + 4a^2 - 3a^2 - 6ab - 3b^2}{12} =$$

$$= \frac{b^2 - 2ab + a^2}{12} = \frac{(b-a)^2}{12}.$$

These are very important expressions. They are used to test random number generators

$$M(X) = \frac{a+b}{2} \quad (6.3)$$

$$D(X) = \frac{(b-a)^2}{12} \quad (6.4) \quad \sigma(X) = \frac{b-a}{2\sqrt{3}} \quad (6.5)$$

Probability that a random variable falls into a segment (α, β) ,
belonging to the segment $[a, b]$,

expressed by the formula

$$P(\alpha < x < \beta) = \int_{\alpha}^{\beta} f(x) dx = \frac{\beta - \alpha}{b - a} \quad (6.6)$$

Uniform distribution occurs when, according to the experimental conditions, the value X takes values in the final segment $[a, b]$.

All values from this segment are possible with equal probability.

For example, the time a passenger waits for transport running at a certain interval; error in rounding a number to an integer;

However, the main application of the theory of uniform distribution is testing random number generators.

Example 6.1

It is known that the transmitter can start working at any time between 12 and 14 hours. What is the probability that you will have to wait no more than 15 minutes for the transmission to start?

Solution.

Let X be the start time of the transmitter in hours.

Since transmission can begin at any moment between 12 and 14 hours and all moments are equally possible, we should assume that X is a random variable uniformly distributed in the segment $[12; 14]$.

Then the distribution density function will take the form:

$$f(x) = \begin{cases} 0, & \text{at } x < 12; \\ \frac{1}{14-12} = 0.5, & \text{at } 12 \leq x \leq 14; \\ 0, & \text{at } x > 14. \end{cases}$$

We find the desired probability using the formula **(6.6)**

$$P(12 < X < 12.25) = \int_{12}^{12.25} 0.5 dx = 0.125$$

It is taken into account that 15 minutes = 0.25 hours.